Figure 2.15 Account balances after completion of step 8

2.8 Smart contract design considerations

Smart contracts are immutable code, like the hardware integrated circuit chips in laptops, smartphones, and computers. In hardware chips, the code is etched in the silicon circuits. Similarly, when smart contracts are deployed, their code is final and cannot be updated unless special provisions or escape hatches are built in. You'll learn about these escape methods in later chapters.

We've also seen that smart contracts are shared with all the stakeholders so that they can independently verify, validate, and reach consensus on the transactions to be recorded in the distributed immutable ledger of the blockchain. Therefore, you must design and test a smart contract thoroughly before deploying it for production use. Though a smart contract is a software module, you cannot change its contents (unlike the weekly updates you get for many apps and operating systems). All these characteristics of a smart contract demand careful design as a precoding step and thorough testing as a postcoding step.

The smart contract is the good, the bad, and the ugly of blockchain technology. It is a powerful feature, but improper design and coding of a smart contract have resulted in significant failure. The Decentralized Autonomous